Kaleb Harris

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SUMMARY

Aspiring Mechanical Engineer at Texas A&M with dual competency in CAD/EDA design and embedded systems, specializing in autonomous systems and robotics. Experienced leading interdisciplinary teams to deliver complex engineering projects from concept to competition-ready hardware.

EDUCATION

Texas A&M | College Station, TX

Northeast Lakeview College | San Antonio, TX — A.S. Engineering | GPA: 3.5 | Expected May 2026

TECHNICAL SKILLS

- **Design & Modeling:** Fusion 360 (3D Assemblies), AutoCAD, MATLAB/Simulink
- **Embedded Systems:** Python, C++, Arduino IDE, ArduPilot/ArduCopter, Microcontrollers (Arduino/Raspberry Pi), PLCs, Git
- **Manufacturing & Testing:** Prototyping, SHM, Autonomous Systems, Data Analysis (CSV/Excel)

ENGINEERING PROJECTS

DIRT Rover — Autonomous Soil Quality Research (TURTLE Robotics, Texas A&M) | Jan 2026 – Present

- Leading a 3-person software team developing autonomous rover systems integrating a deep spectroscopy probe for real-time soil quality measurement, with full ownership of architecture decisions, task delegation, and sprint planning.
- Building rover coding, data handling, and automation pipelines to support remote operation and real-time sensor data acquisition; coordinating software integration with mechanical and electrical subsystems across varying terrain conditions.
- Iterating system design in alignment with TURTLE Robotics' goals of advancing A&M agricultural research through accurate, field-deployable sensor technology.

Fixed-Wing UAV Design and Environmental Data Acquisition Using Additive Manufacturing (NLC STEM Club) | Jan 2026 – Present

- Directing an interdisciplinary 4-team effort to design and build a hybrid solar-powered quadcopter (250mm frame, ~4–6 lbs) running ArduCopter/ArduPilot, targeting 15–25 min flight time with onboard VOC, CO2, temperature, and humidity sensors.
- Managing a 10-week development cycle across Mechanical, Electronics, Software, and Documentation teams from initial design through systems integration, flight testing, and final research publication.

Piezoelectric Structural Health Monitoring (SHM) System (NLC STEM Club) | Oct 2025

- Co-developed a comparative study using piezoelectric transducers and an Arduino Uno DAQ system to detect real-time structural damage on anisotropic (Cedar) and isotropic (3D-printed PLA) bridge models.
- Engineered a web-based Data Acquisition (DAQ) system for remote test initiation and real-time visualization of sensor harmonics.
- Identified a 15-second pre-failure cascade during load-to-failure testing, validating the system for predictive failure analysis; documented an 850 mV peak response in PLA versus 400 mV in wood.

LEADERSHIP EXPERIENCE

TURTLE Robotics (Texas A&M) | College Station, TX | ***Software Team Lead — DIRT Project*** | Jan 2026 – Present

- Directing software development strategy for an autonomous agricultural rover, managing a 3-person team through full-cycle development including requirements gathering, subsystem integration, and iterative testing.
- Serving as the primary liaison between software and cross-functional engineering teams, ensuring alignment on technical requirements and project milestones within TURTLE Robotics' research framework.

STEM Club (NLC) | San Antonio, TX | ***Founder & President*** | Jan 2024 – Present

- Founded the first campus-wide STEM org; grew to 90+ members through weekly technical workshops, industry guest lectures, and coordinating interdisciplinary engineering projects including UAV and SHM research initiatives.

Schertz Family YMCA | Universal City, TX | ***Lifeguard / Lead Technical Trainer*** | Mar 2022 – Present

- Directs emergency response training ensuring 100% staff safety compliance; communicates technical safety procedures to 100+ patrons daily.